Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Solving Systems Using Elimination and Substitution

Name: ____________
Date: ____________

Quiz (Habanero)

Q1. Solve the system: $x + 2y = 7$, $3x - 2y = 5$. What is x?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q2. A weather front moves into an area, causing the temperature to drop 3°F per hour. If the initial temperature is 75°F, and t represents time in hours, which equation models the temperature T?
(A) $T = 75 - 3t$
(B) $T = 75 + 3t$
(C) $T = 3t - 75$
(D) $T = 3t + 75$
(E) $T = -3t + 75$

Q3. What is the solution to the system: $y = 2x - 1$, $y = -x + 3$?
(A) (1, 1)
(B) (2, 3)
(C) (1.5, 2)
(D) (2.5, 4)
(E) No solution

Q4. Solve the system: $x + y = 4$, $2x - 2y = 4$. What is y?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4

Q5. A drought-stricken area receives rainfall at a rate of 2 inches per day. If the total rainfall R is 10 inches after t days, which equation models this situation?
(A) $R = 2t + 10$
(B) $R = 2t - 10$
(C) $R = 10 - 2t$
(D) $R = 2t$
(E) $R = 10 + 2t$

Q6. Determine the number of solutions to the system: $x - 2y = 3$, $2x - 4y = 6$.
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions

Q7. What is the solution to the system: $y = -x + 2$, $y = x - 2$?
(A) (0, 0)
(B) (2, 0)
(C) (0, 2)
(D) No solution
(E) Infinitely many solutions

Q8. Solve the system: $2x + 3y = 7$, $x - 2y = -3$. What is y?
(A) 1
(B) 2
(C) -1
(D) -2
(E) 0

Open Ended Questions (FRQ)

Q9. A forest's ecosystem is affected by the amount of rainfall and temperature. Suppose the amount of rainfall R and temperature T are related by the system: $R = 2T - 5$, $T = R + 2$. Solve for R and T.

Q10. An environmental study examines the relationship between the amount of pollutants P in a river and the amount of oxygen O in the water. The system of equations is: $P = 3O - 12$, $2P + O = 10$. Solve for P and O.

Chef's Tips

- Provide detailed solutions for FRQ questions.
- For MQ questions, ensure students show work and explain their reasoning.
- For word problems, encourage students to use real-world applications and examples.